



Unit 1: Introduction to Microeconomics

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1



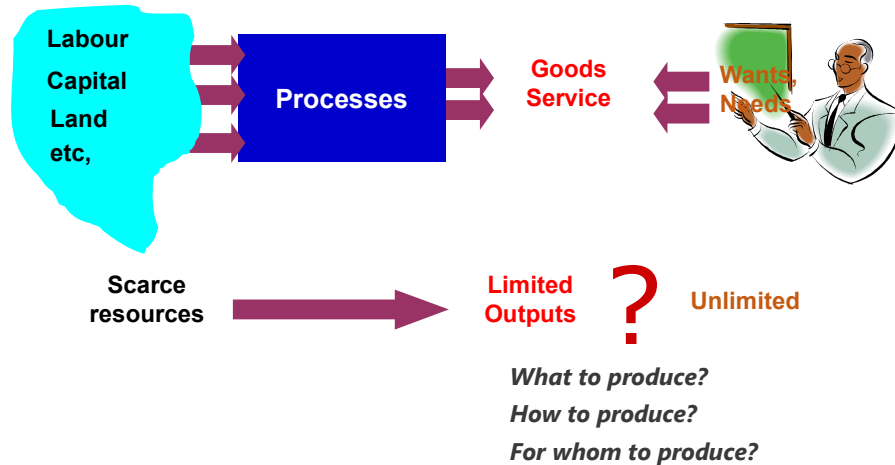
Contents

- 1.1 Definition of Economics
- 1.2 Ten principles of Economics
- 1.3 Thinking like an economist
- 1.4 Economic models
- 1.5 Microeconomics and macroeconomics
- 1.6 Positive versus normative statements
- 1.7 Why economics disagree?

2



1.1 What is an economics?



3



1.1 What is an Economics?

- **Scarcity**. . . means that society has limited resources and therefore cannot produce all the goods and services people wish to have.
- The management of society's resources is important because resources are scarce.
- **Economics** is the study of how society manages its scarce resources.

4



Economic Study



- Economists study how people make decisions
 - How much they work
 - What they buy
 - How much they save
 - How they invest their savings
- Economists also study how people interact such as buyers and sellers
 - Price determination
- Economists also analyze forces and trends that affect the economy as a whole
 - Growth in average income
 - The rate of price increase.

5



1.2 Ten Principles of Economics



- There is no mystery to what an “economy” is.
- It’s a group people interacting with one another as they go about their lives.
- **Four principles of individual decision making:**
 1. People face tradeoffs
 2. The cost of something is what you give up to get it
 3. Rational people think at the margin
 4. People respond to incentives

6



Principle 1: People Face Tradeoffs

- To get something we like we usually have to give up something we don't like.
 - A student and her(his) time:
 - Studying vs. napping or cycling
 - Society's tradeoffs:
 - Guns vs. Butter
 - Clean environment vs. higher income
 - Efficiency vs. Equity
 - Efficiency: Society getting the most it can from its scarce resources.
 - Equity: Distributing economic prosperity fairly among the members of society.

7



Principle 2: The Cost of Something is What You Give Up

- Making decisions requires comparing the *costs* and *benefits* of alternative courses of actions.
 - To go to university or not to go?
- **Opportunity cost:** Whatever must be given up to obtain some item.

8



Principle 3: Rational People Think at the Margin

- **Rational people**
 - systematically and purposefully do the best they can to achieve their objectives.
- Make decisions by evaluating costs and benefits of **marginal changes** – incremental adjustments to an existing plan.
- Individuals and firms can make better decisions by thinking at the margin.
 - By comparing the *marginal benefits* (MB) with the associated *marginal costs* (MC) of a decision.

9



Your Decision

- **Which should be used: fluorescent lights or LED lights?**
 - Fluorescent lights cost 20,000 VND per unit, with a lifespan of 15,000 hours and an electricity consumption of 38W per hour.
 - LED lights cost 100,000 VND per unit, with a lifespan of 25,000 hours and an electricity consumption of 20W per hour.
 - The electricity price is 2,000 VND per kWh. Both types of lights provide the same luminous flux."

10



Do You want to Use Electric Motorbike?



Electric Motorbike	Gasoline Motorbike
Motorbike lifespan: 5 years	
Monthly distance traveled: 500 km	
E-bike (with battery): 21,000,000 VND	Gasoline motorbike price: 18,000,000 VND
Electricity charging cost: 3,400 VND/kWh	RON 95 gasoline price: 21,000 VND/liter
Electricity consumption: 3 kWh/100 km	Fuel consumption: 2.0 liters/100 km
Maintenance cost: 400,000 VND/year	Maintenance cost: 1,100,000 VND/year
Salvage value: 2,000,000 VND	Salvage value: 5,000,000 VND
Charging time per session: 6 hours Travel distance per charge: 200 km Probability of daytime charging: 10% Average wage: 50,000 VND/hour	Refueling time: negligible

11



Principle 4: People Respond to Incentive

- **Incentive:** something that induces a person to act, *i.e.* the prospect of a reward or punishment.
- Rational people respond to incentives.
- Examples:
 - When oil prices rise, consumers buy more electric cars and fewer gas guzzling SUVs.
 - When cigarette taxes increase, teen smoking falls.

12



How People Interact

- The first four principles discussed *how individuals make decisions*.
- The next three principles concern *how people interact with one another*.
 5. Trade can make everyone better off
 6. Markets are usually a good way to organize economic activity
 7. Governments can improve market outcomes

13



Principle 5: Trade can Make Everyone Better Off

- Rather than being self-sufficient, people can specialize in producing one good or service and exchange it for other goods.
- Countries also benefit from trade & specialization:
 - Get a better price abroad for goods they produce
 - Buy other goods more cheaply from abroad than could be produced at home

14



Principle 6: Markets are Usually a Good Way to Organize Economic Activity

- Market economy: An economy that allocates resources through the decentralized decisions of many firms and households as they interact in markets for goods and services.
- Firms decide whom to hire and what to make.
- Households decide which firms to work for and what to buy with their incomes.

15



Principle 7: Governments can Improve Market Outcomes

- Market failure: A solution in which a market left on its own fails to allocate resources efficiently.
 - Externality: The impact of one person's actions on the well-being of a bystander.
 - Market power: The ability of a single economic actor (or small group of actors) to have a substantial influence on market prices.
- In such cases, public policy may [promote efficiency](#).
- Govt. may alter market outcome to [promote equity](#).
 - If the market's distribution of economic well-being is not desirable, tax or welfare policies can change how the economic "pie" is divided.

16



How The Economy as a Whole Works

- The last three principles concern the *workings of the economy as a whole*:
 8. A country's standard of living depends on its ability to produce goods and services;
 9. Prices rise when the government prints too much money;
 10. Society faces a short-run tradeoff between inflation and unemployment.

17



Principle 8: A Country's Standard of Living Depends on its Ability to Produce Goods and Services

- **Standard of Living** may be measured in different ways (e.g. personal income or total market value of a nation's production.)
 - Differences in standard of living between countries or even provinces is attributable to the *productivity* of the country or province.
- **Productivity**: The amount of goods and services produced from each hour of a worker's time.

Productivity => Standard of Living

18



Principle 9: Prices Rise when the Government Prints Too Much Money

- **Inflation:** increases in the general level of prices.
- In the long run, inflation is almost always caused by excessive growth in the quantity of money, which causes the value of money to fall.
- The faster the govt creates money, the greater the inflation rate.



19



Principle 10: Society Faces a Short-Run Tradeoff Between Inflation and Unemployment.

- In the short-run (1 – 2 years), many economic policies push inflation and unemployment in opposite directions.
- Other factors can make this tradeoff more or less favorable, but the tradeoff is always present.
- **Phillips curve:** A curve that shows the short-run tradeoff between inflation and unemployment.



20



1.3 Thinking Like an Economist

Every field of study has its own terminology

- Mathematics
 - integrals ❖ axioms ❖ vector spaces
- Psychology
 - ego ❖ id ❖ cognitive dissonance
- Law
 - promissory ❖ estoppels ❖ torts ❖ venues
- Economics
 - supply ❖ opportunity cost ❖ elasticity ❖ consumer surplus ❖ demand ❖ comparative advantage ❖ deadweight loss

21



Thinking Like an Economist

- Economics trains you to. . . .
 - Think in terms of alternatives.
 - Evaluate the cost of individual and social choices.
 - Examine and understand how certain events and issues are related.

22



The Economist as a Scientist

The economic way of thinking . . .

- Involves thinking analytically and objectively.
- Economists...
 - Devise theories
 - Collect data
 - Analyze the data to verify or refute their theories
- Economics makes use of the *scientific method*.



23



The Scientific Method: observation, theory, and more observation

- Uses abstract models to help explain how a complex, real world operates.
- Develops theories, collects, and analyzes data to evaluate the theories.



24



The Role of Assumptions

- Economists make assumptions in order to make the world easier to understand.
- The art in scientific thinking is deciding which assumptions to make.
- Economists use different assumptions to answer different questions.

“Ceteris paribus”

25



1.4 Economic Models

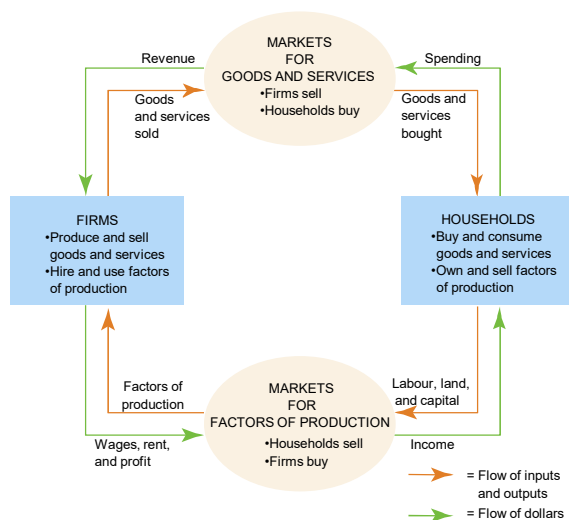
- Economists use models to simplify reality in order to improve our understanding of the world
- Two of the most basic economic models include:
 - The *Circular Flow Diagram*
 - The *Production Possibilities Frontier*

26



First Model: The Circular-Flow Diagram

- The *circular-flow diagram* is a visual model of the economy that shows how dollars flow through markets among households and firms.



27



- Firms**
 - Produce and sell goods and services
 - Hire and use factors of production
- Households**
 - Buy and consume goods and services
 - Own and sell factors of production
- Markets for Goods and Services**
 - Firms sell
 - Households buy
- Markets for Factors of Production**
 - Households sell
 - Firms buy
- Factors of Production**
 - Inputs used to produce goods and services
 - Land, labor, and capital

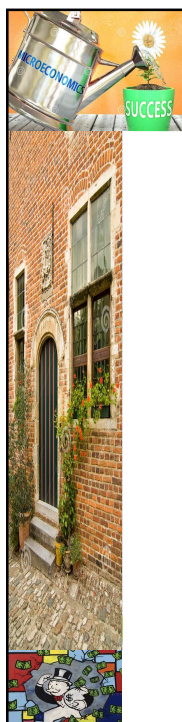
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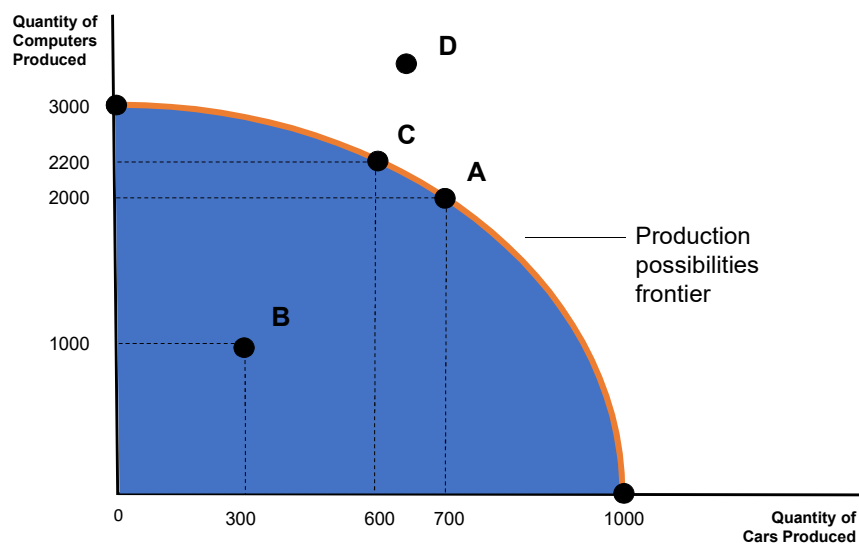
Second Model: The Production Possibilities Frontier

- The *production possibilities frontier* is a graph that shows the combinations of output that the economy can possibly produce given the available factors of production and the available production technology.

29



The Production Possibilities Frontier



30



1.5 Microeconomics and Macroeconomics

- *Microeconomics* focuses on the individual parts of the economy.
 - How households and firms make decisions and how they interact in specific markets
- *Macroeconomics* looks at the economy as a whole.
 - Economy-wide phenomena, including inflation, unemployment, and economic growth



31



1.6 Positive Versus Normative Statements

- *Positive statements* are statements that attempt to describe the world as it is.
 - Called descriptive analysis
- *Normative statements* are statements about how the world should be.
 - Called prescriptive analysis



32



Positive Versus Normative Statements

- Positive or Normative Statements?

- An increase in the minimum wage will cause a decrease in employment among the least-skilled.

POSITIVE

- Higher government budget deficits will cause interest rates to increase.

POSITIVE

33



Positive Versus Normative Statements

- Positive or Normative Statements?

- The income gains from a higher minimum wage are worth more than any slight reductions in employment.

NORMATIVE

- State governments should be allowed to collect from tobacco companies the costs of treating smoking-related illnesses among the poor.

NORMATIVE

34



1.7 Why Economists Disagree?

- They may disagree about the validity of alternative positive theories about how the world works.
- They may have different values and, therefore, different normative views about what policy should try to accomplish.

